CELLULAR GENETICS

Module map

- Module 08: Cellular Genetics
- Connected lab: lab-08_cellular-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 08

Cellular Genetics

1. Chromosomes and cell cycle
2. Mitosis for growth and repair
3. Meiosis for gametes
4. Variation from recombination and assortment
5. Errors and genetic consequences

Lab evidence

lab-08_cellular-genetics.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

CELLULAR GENETICS

Learning objectives

- Relate chromosomes to DNA organization.
- Compare the purpose and products of mitosis and meiosis.
- Explain how meiosis reduces chromosome number.
- Describe how crossing over and independent assortment increase variation.
- Interpret chromosome number errors at an introductory level.

OBJECTIVE 1
Relate chromosomes to DNA organization.

OBJECTIVE 2
Compare the purpose and products of mitosis and meiosis.

OBJECTIVE 3
Explain how meiosis reduces chromosome number.

OBJECTIVE 4
Describe how crossing over and independent assortment increase variation.

OBJECTIVE 5
Interpret chromosome number errors at an introductory level.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

CELLULAR GENETICS

Topic sequence

- Chromosomes and cell cycle: Chromosomes organize DNA so cells can divide accurately.
- Mitosis for growth and repair: Mitosis produces genetically identical daughter cells for growth and repair.
- Meiosis for gametes: Meiosis produces haploid gametes for sexual reproduction.
- Variation from recombination and assortment: Crossing over and independent assortment create genetic variation.
- Errors and genetic consequences: Nondisjunction and other errors can change chromosome number.

01

Chromosomes and cell cycle

Chromosomes organize DNA so cells can divide accurately.

02

Mitosis for growth and repair

Mitosis produces genetically identical daughter cells for growth and repair.

03

Meiosis for gametes

Meiosis produces haploid gametes for sexual reproduction.

04

Variation from recombination and assortment

Crossing over and independent assortment create genetic variation.

05

Errors and genetic consequences

Nondisjunction and other errors can change chromosome number.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

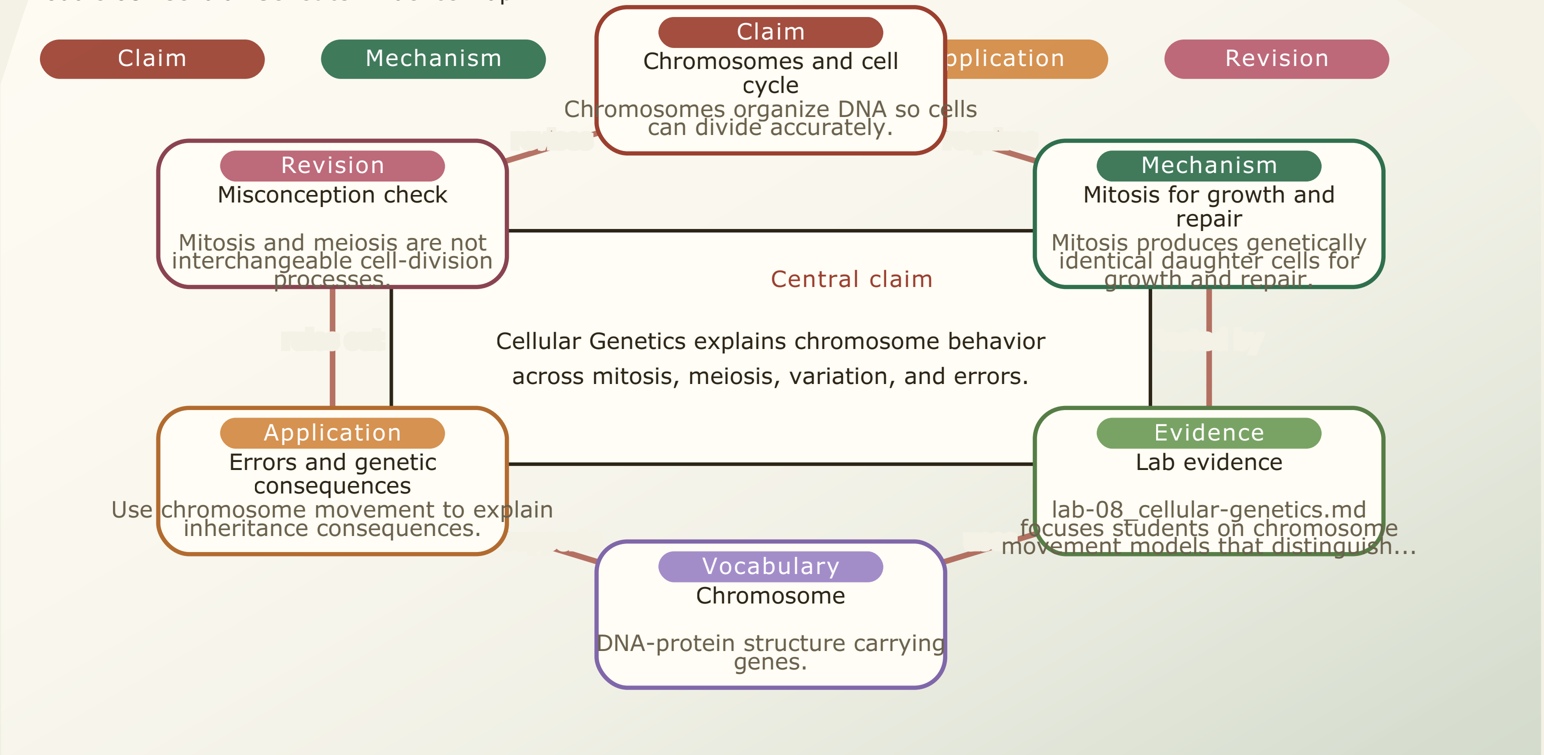
CELLULAR GENETICS

Module 08: Cellular Genetics Evidence Map

- Module focus: Cellular Genetics
- Central claim: Cellular Genetics explains chromosome behavior across mitosis, meiosis, variation, and errors.
- Key nodes to connect: Chromosomes and cell cycle, Mitosis for growth and repair, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-08_cellular-genetics.md supplies an evidence surface for the map.

Module 08 / Concept Map

Module 08: Cellular Genetics Evidence Map



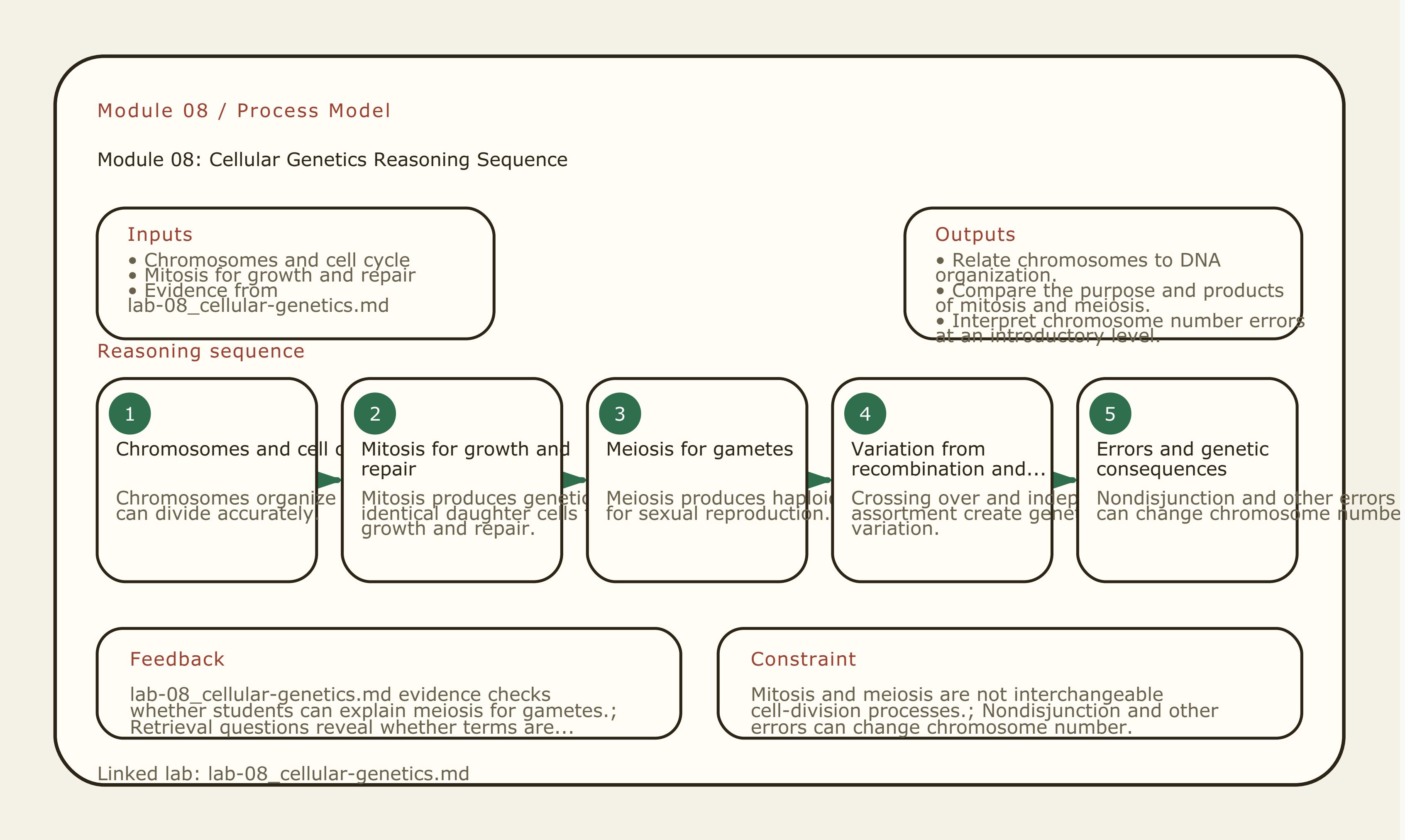
Linked lab: lab-08_cellular-genetics.md / Generated from explicit module.toml visual schema

Teaching note: Ask students to explain one edge in their own words before moving on.

CELLULAR GENETICS

Module 08: Cellular Genetics Reasoning Sequence

- Module focus: Cellular Genetics
- Trace the model: Chromosomes and cell cycle -> Mitosis for growth and repair -> Meiosis for gametes -> Variation from recombination and assortment
- Inputs become outputs: Chromosomes and cell cycle, Mitosis for growth and repair -> Relate chromosomes to DNA organization., Compare the purpose and products of mitosis and meiosis.
- Feedback check: lab-08_cellular-genetics.md evidence checks whether students can explain meiosis for gametes.
- Lab connection: lab-08_cellular-genetics.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

CELLULAR GENETICS

Terms as evidence handles

- Chromosome: DNA-protein structure carrying genes.
- Cell cycle: Ordered sequence of cell growth, DNA replication, and division.
- Mitosis: Nuclear division producing identical daughter nuclei.
- Meiosis: Division process producing haploid gametes.
- Gamete: Haploid reproductive cell.
- Crossing over: Exchange of DNA between homologous chromosomes.

CHROMOSOME
DNA-protein structure carrying genes.

CELL CYCLE
Ordered sequence of cell growth, DNA replication, and division.

MITOSIS
Nuclear division producing identical daughter nuclei.

MEIOSIS
Division process producing haploid gametes.

GAMETE
Haploid reproductive cell.

CROSSING OVER
Exchange of DNA between homologous chromosomes.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-08_cellular-genetics.md
- Lab evidence should test or illustrate: Meiosis for gametes
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Chromosomes and cell cycle

EVIDENCE
lab-08_cellular-genetics.md

CLAIM
Relate chromosomes to DNA organization.

Teaching note: Make the lab connection explicit before students encounter the activity.

CELLULAR GENETICS

Contrast check

- Do not stop at: Chromosomes organize DNA so cells can divide accurately.
- Stronger explanation: Nondisjunction and other errors can change chromosome number.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Chromosomes organize DNA so cells can divide accurately.

DEEPER
Nondisjunction and other errors can change chromosome number.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

CELLULAR GENETICS

Module 08: Cellular Genetics Retrieval and Lab Check

- Module focus: Cellular Genetics
- Retrieval prompt: Why do cells package DNA into chromosomes?
- Answer check: Relate chromosomes to DNA organization.
- Required terms: Chromosome, Cell cycle, Mitosis
- Lab connection: lab-08_cellular-genetics.md; lab-08_cellular-genetics.md supplies evidence for chromosome movement models that distinguish mitosis from meiosis.

Module 08 / Retrieval Card

Module 08: Cellular Genetics Retrieval and Lab Check

Routine

Cover notes -> answer aloud -> cite evidence -> revise

1 Cover notes | 2 Answer aloud | 3 Cite evidence | 4 Revise

Why do cells package DNA into chromosomes?

Check: Relate chromosomes to DNA organization.

What is the purpose of mitosis?

Check: Compare the purpose and products of mitosis and meiosis.

What is the purpose of meiosis?

Check: Explain how meiosis reduces chromosome number.

How do the products of mitosis and meiosis differ?

Check: Describe how crossing over and independent assortment increase variation.

Terms to use

Chromosome, Cell cycle, Mitosis, Meiosis, Gamete

Lab connection

lab-08_cellular-genetics.md supplies evidence for chromosome movement models that distinguish mitosis from...

Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. What is the main product of mitosis?
- 2. Meiosis reduces the chromosome number by using:
- 3. Crossing over and independent assortment both act to:
- 4. If chromosomes fail to separate correctly during meiosis (nondisjunction), a resulting gamete may have:

QUESTION 1

Key: B

What is the main product of mitosis?

QUESTION 2

Key: D

Meiosis reduces the chromosome number by using:

QUESTION 3

Key: A

Crossing over and independent assortment both act to:

QUESTION 4

Key: C

If chromosomes fail to separate correctly during meiosis (nondisjunction), a resulting gamete may have:

Teaching note: Use the quiz as formative feedback. Answer key: 1=B, 2=D, 3=A, 4=C

CELLULAR GENETICS

Synthesis and exit ticket

- Exit claim: explain chromosomes and cell cycle using evidence from lab-08_cellular-genetics.md.
- Must include: Chromosome, Cell cycle, Mitosis.
- Revision prompt: what evidence would change your explanation?

CLAIM
Chromosomes and cell cycle

EVIDENCE
lab-08_cellular-genetics.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.